Open WebUI

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For users using large AI model images, the environment has already been set up. All setup steps in the tutorial are unnecessary; jump directly to the running step.

> **Demo Environment**

Development Board: Raspberry Pi

SD (TF) Card: 64GB (16GB or more; the larger the card, the more models you can experiment with)

```
Raspberry Pi 5 (16GB RAM): Runs models with 14B or fewer parameters
Raspberry Pi 5 (8GB RAM): Runs models with 8B or fewer parameters
Raspberry Pi 5 (4GB RAM): Runs models with 3B or fewer parameters
Raspberry Pi 5 (2GB RAM): Runs models with 1B or fewer parameters
```

Open WebUI is an open source project designed to provide a simple and easy-to-use user interface (UI) for managing and monitoring open source software and services.

```
When using Open WebUI, you may experience unresponsiveness or timeouts. Try
restarting Open WebUI or switching to the Ollama tool to run the model.
```

Environment Requirements

Host and Conda Installation of Open WebUI: Requires Node.js 20.10 or higher, Python 3.11:

Environment Setup Method	Difficulty (Relative)
Host	High
Conda	Medium
Docker	Low

This tutorial demonstrates installing Open WebUI using Docker.

Docker Setup

Official Docker Installation

If Docker is not installed, you can use a script to install Docker in one click.

```
The Raspberry Pi 5 image we provide already has Docker installed; no need to
install it yourself!

Users in China may not be able to install Docker through official methods. We
recommend using Alibaba Cloud to install Docker or using our image directly.
```

- Updating the Local Package List.

```
sudo apt update
```

- Upgrading installed packages.

```
sudo apt upgrade
```

- Download and run the script.

Download the get-docker.sh file and save it in the current directory.

```
sudo apt install curl
```

```
curl -fsSL https://get.docker.com -o get-docker.sh
```

Run the get-docker.sh script file with sudo privileges.

```
sudo sh get-docker.sh
```

Installing Docker on Alibaba Cloud

If you can't install it yourself, please use the image we provide.

- Update the local package list.

```
sudo apt update
```

- Install prerequisite software.

```
sudo apt install apt-transport-https ca-certificates curl gnupg2 lsb-release
software-properties-common
```

- Add the GPG key of the software repository.

```
curl -fsSL https://mirrors.aliyun.com/docker-ce/linux/debian/gpg | sudo gpg --
dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg
```

- Add Alibaba Cloud image software source.

```
echo "deb [arch=arm64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
https://mirrors.aliyun.com/docker-ce/linux/debian bookworm stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null
```

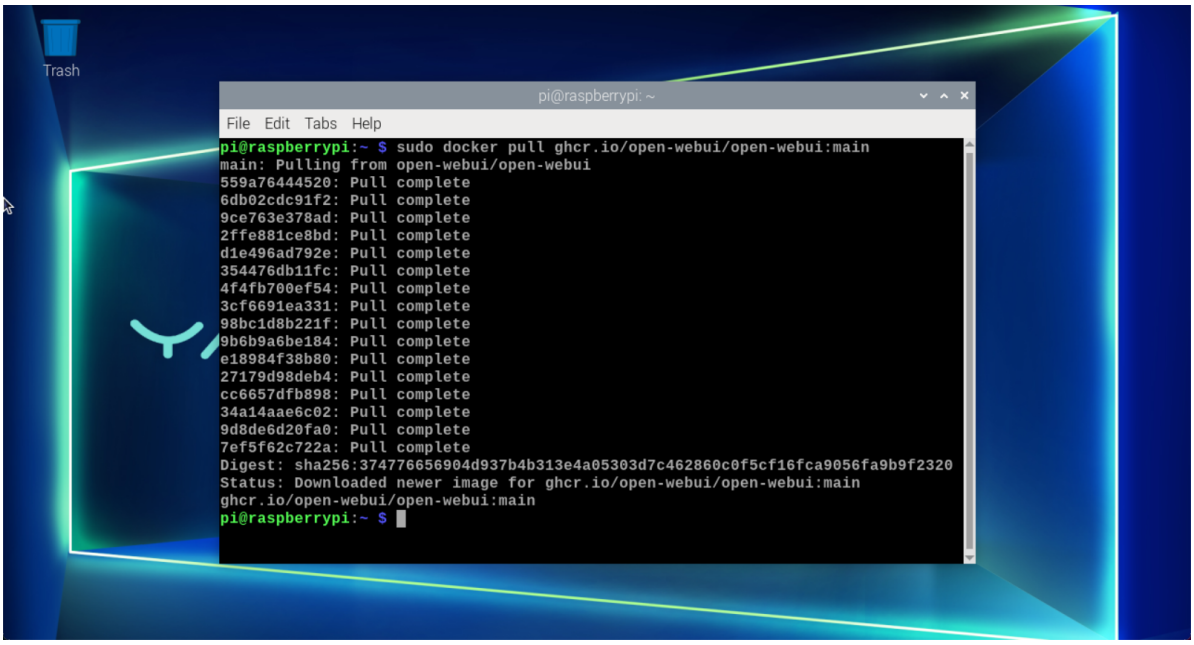
- Install Docker.

```
sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

Open WebUI Installation

For systems with Docker installed, you can directly enter the following command in the terminal:

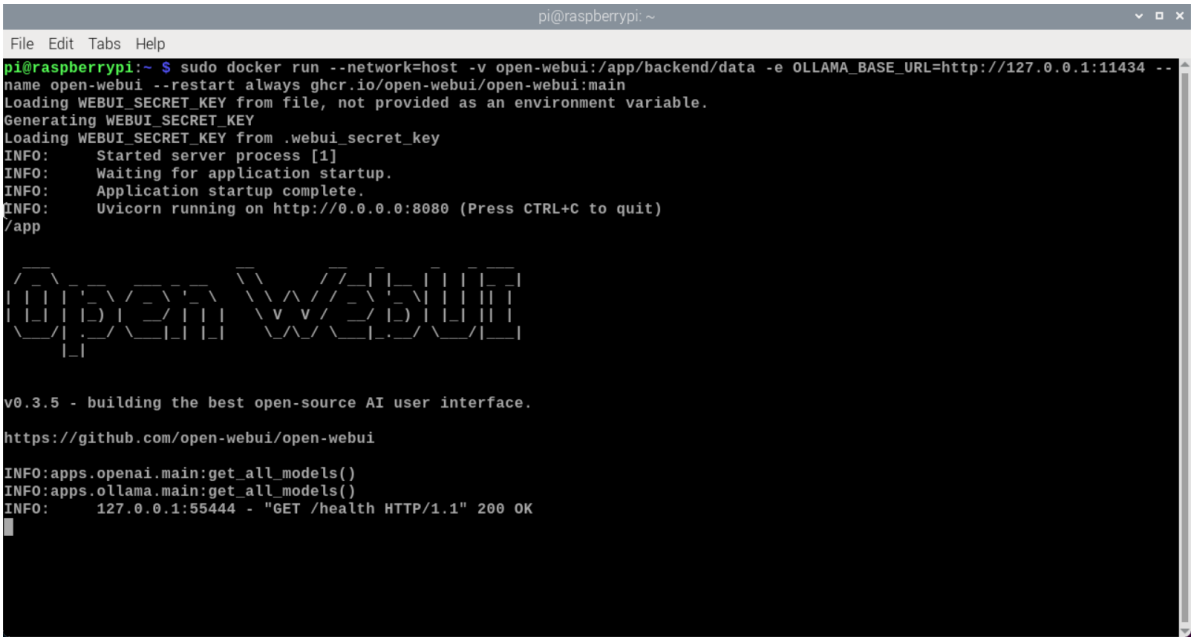
```
sudo docker pull ghcr.io/open-webui/open-webui:main
```



Open the WebUI and run it

Enter the following command in the terminal to start the specified Docker:

```
sudo docker run --network=host -v open-webui:/app/backend/data -e
OLLAMA_BASE_URL=http://127.0.0.1:11434 --name open-webui --restart always
ghcr.io/open-webui/open-webui:main
```



After successful startup, use the following URL to access the browser:

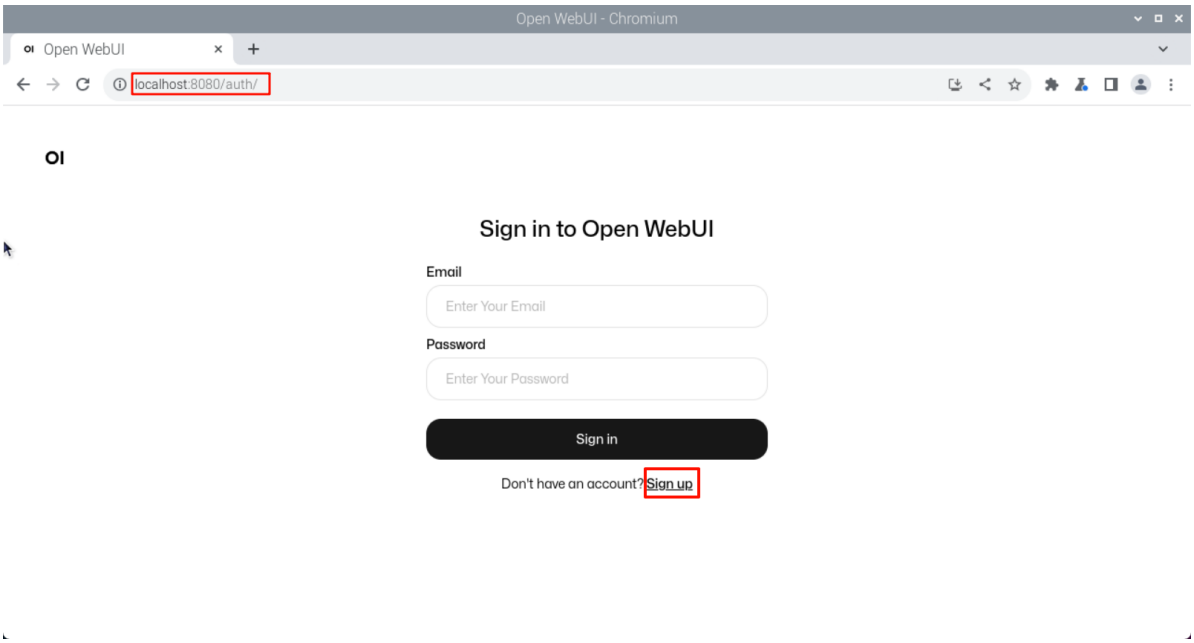
```
http://localhost:8080/
```

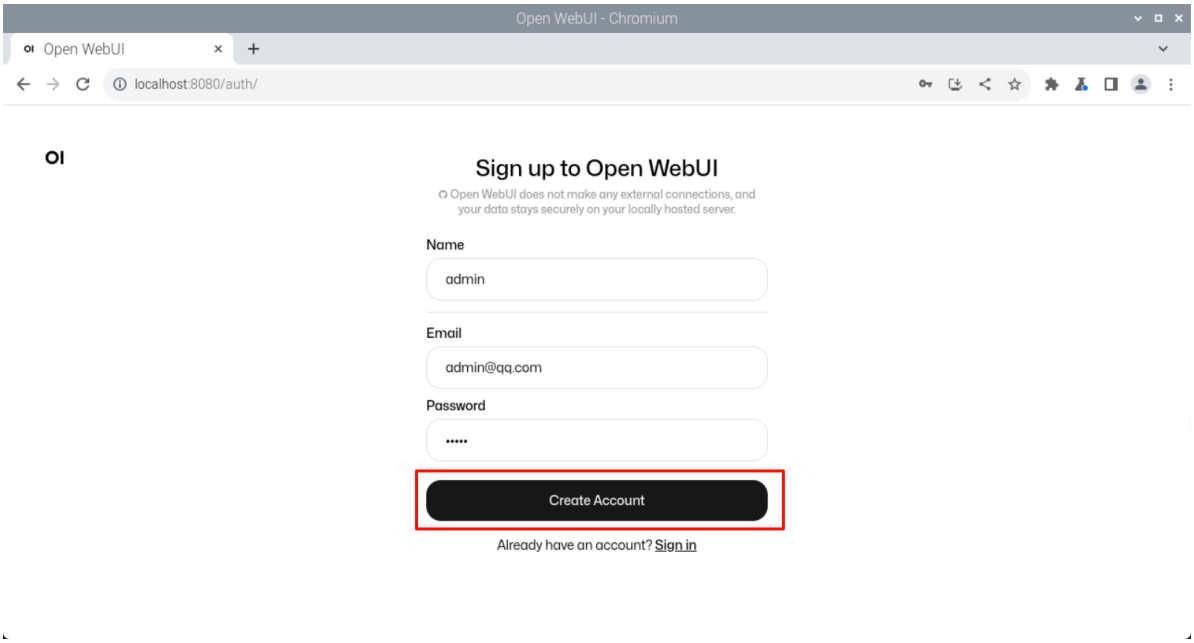
Administrator Account

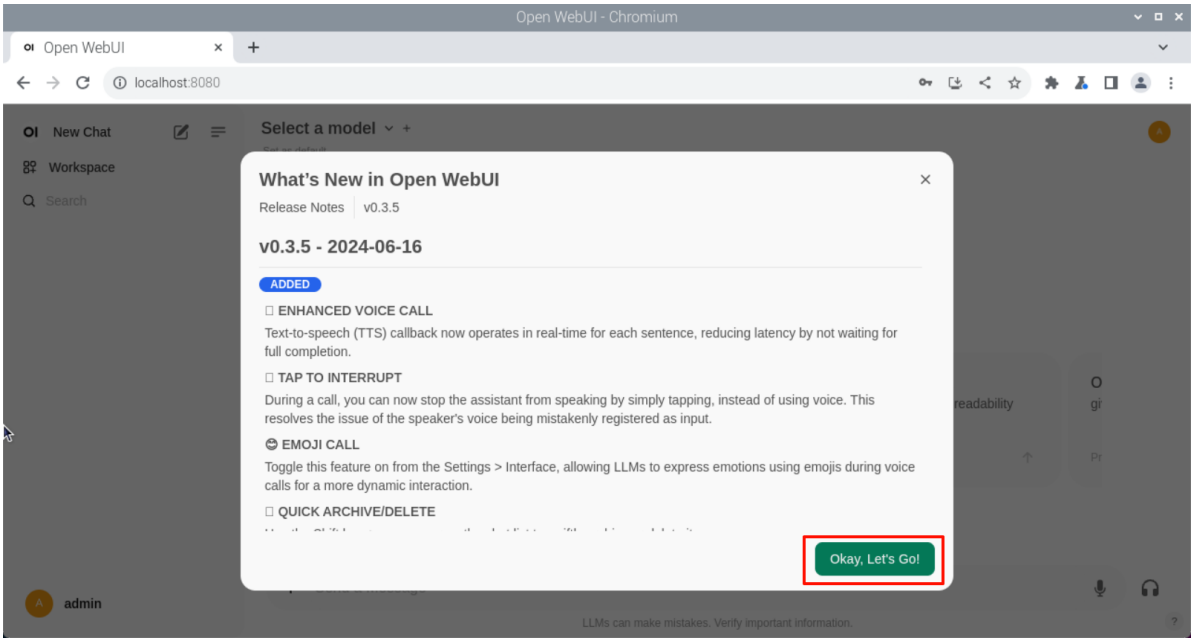
For first-time users, you'll need to register an administrator account. Fill in the required information as needed!

```
Since all content in our image has been set up and tested, you can log in
directly using our registered account:
Username: yahboom
Email: yahboom@163.com
Password: yahboom
```

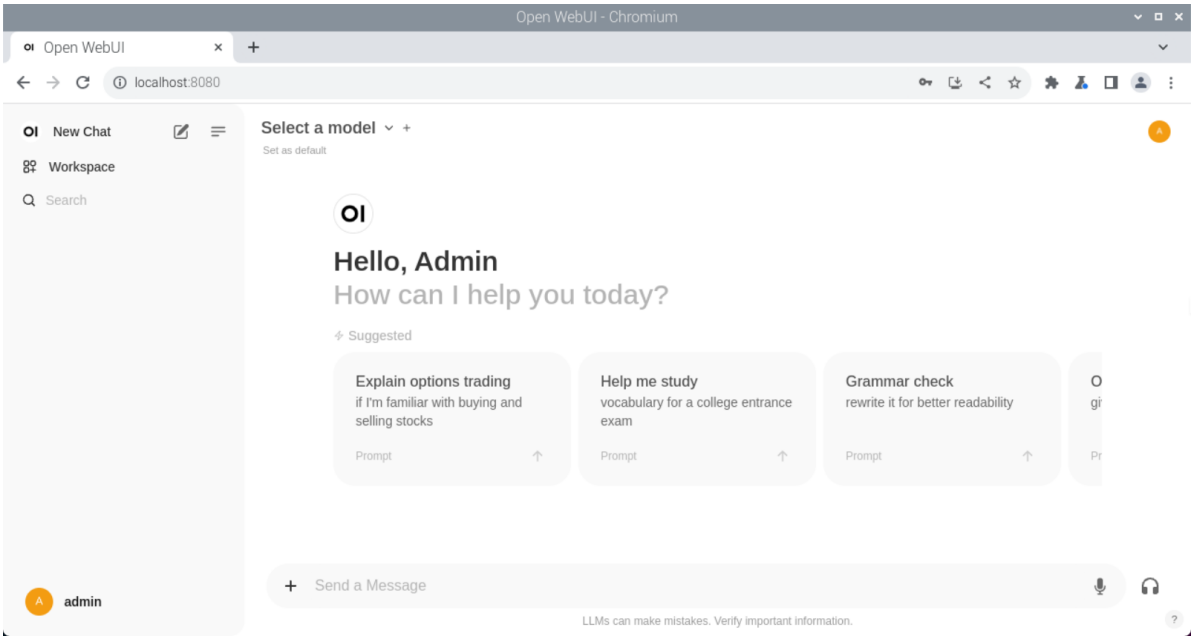
Register and Login







User Interface



Model Dialog

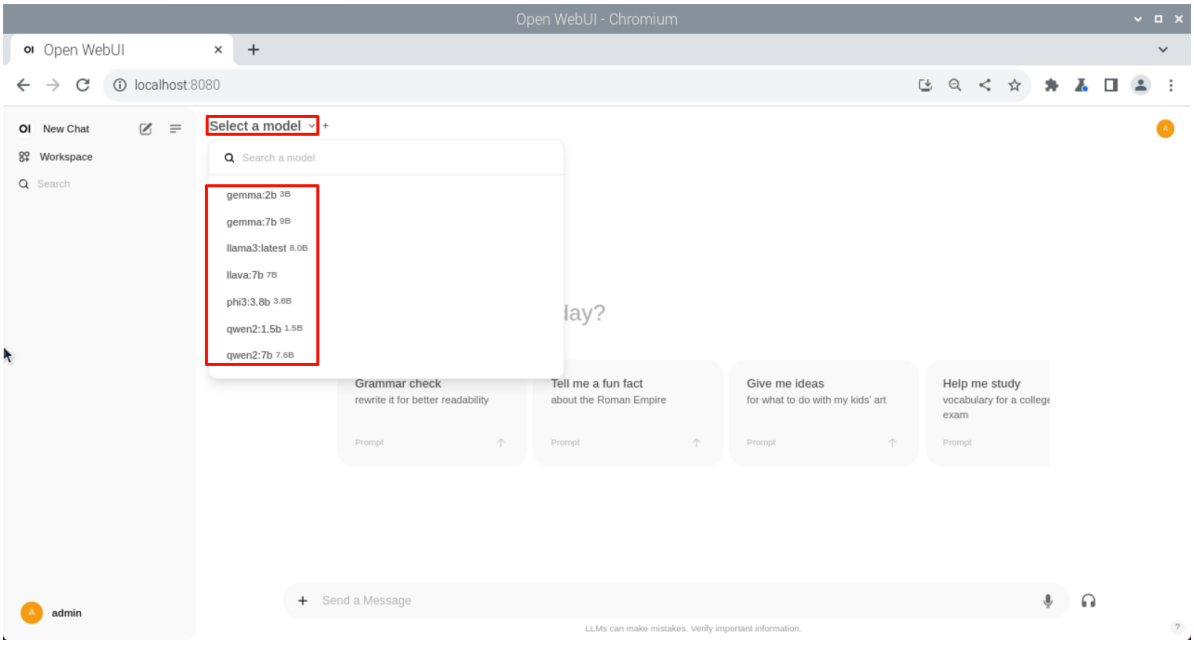
Using Open WebUI for dialog is slower than using the Ollama tool directly, and may even result in timeouts and service connection failures. This is related to the Raspberry Pi 5's memory and is unavoidable!

```
Expert users can switch to another Linux environment to set up Ollama and Open
WebUI for dialog.
```

Switching Models

Click `Select a model` to select a specific model for dialog.

```
Models pulled using Ollama will automatically be added to the Open WebUI model
selection. Refresh the page to see the new model!
```



FAQ

Closing the Open WebUI

Close the automatically started Open WebUI.

- Viewing the running Docker.

```
docker ps
```

- Shut down the running Docker.

```
docker stop [CONTAINER ID]  # Example docker stop 5f42ee9cf784
```

- View stopped containers.

```
docker ps -a
```

- Cleaning up stopped containers.

```
docker rm [CONTAINER ID]    # Example docker rm 5f42ee9cf784
```

Clean up all stopped containers:

```
docker container prune
```

Common Errors

Unable to start Open WebUI

- docker: Error response from daemon: Conflict. The container name "/open-webui" is already in use by container "cfc05c84f8e38b290337e7178c76fd1c49076f94b11ed3d49d9448be72b7f20f". You have to remove (or rename) that container to be able to reuse that name.

Solution: Close Open WebUI and restart it!

Service connection timed out

- Open WebUI: Server Connection Error

Close Open WebUI once and restart it, then ask the question again or use the Ollama tool to run the model and ask the question!